

Making symmetric block cipher meta permutating again

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Abstract

Hi, I'm [heinrichelsigan](#). I'm interested in CSharp, Java, MSSQL, .Net Core, Android and politics, society and the future; currently working as freelancer (one person company) and planning a secure endpoint 2 entpoint chat and looking to collaborate on reviews for other repositories and projects.

1. social media, development and other links:
2. personal tech and political blog [blog.area23.at](#)
3. GitHub repositories [github.com/heinrichelsigan](#)
4. live demo .Net [area23.at/net/](#)
5. StackOverflow [stackoverflow.com/users/12213151/heinrich-elsigan](#)
6. Curriculum vitae [heinrichelsigan.area23.at/cv](#)

1 Thanks to

Normally, thanks to are always at the end of each paper, but the people or organizations I benefited from while trying to make AES strong again are more important than a simple proof of concept showing that it works, except on my raw experimental test form.

1. [bouncycastle.org](#)

2. schneier.com
3. github.com/dotnet
4. github.com/microsoft
5. git.lysator.liu.se/nettle/nettle real easy to read c/c++ code

Go to io.cqrxs.eu/download and choose latest version. Attention, version before March 2025 have a 3-fish over Aes-Engine encryption bug, because 3-fish uses AES default block size and key length and Bouncy Castle Aes-Engine parameters. It works, but settings AES default engine for 3-fish, isn't so serious and please download version after March 30.

2 Documentation online en-/decrypt form

1. Key: When clicking key your entered key will be stored temporary in session.
2. Textbox secret key: Enter your Email address and secret key.
3. Buton Clear: Clear and reset the entire form.
4. Hyperlink Help: Show this help
5. RadioButtonList KeyHashes Choose the hash method to hash your secret key.
6. ImageButton Hash: Clicking will hash your key and display hashed key in textbox.
7. TextBox Hash (readonly): Displays your hashed key.
8. Button "Set Pipe": sets symmetric cipher pipe, dependent only on your entered key
9. Button "Hash Pipe": sets symmetric cipher pipe, dependent primary on calculated hash and secondary on your entered key.
10. DropDown ZipTypes Choose encryption type (Please only GZip or Zip or None)
11. DropDown CipherTypes Choose a symmetric cipher to add it to the symmetric cipher pipe.
12. ImageButton „add algo“: Clicking on will add the in c. DropDown CipherTypes selected symmetric cipher algorithm to CipherPipe.
13. TextBox CipherPipe (readonly): Displays the current Cipher Pipe algorithms.
14. ImageButton "Clear Pipe": Clicking on will clear only the entire Cipher Pipe
15. DropDown EncodingTypes: Choose the final binary to ascii encoder, default is Base64. Beware of using uuencode in Web, because `j` will be interpreted as possible html injection.
16. TextArea Source: paste or enter Text here.
17. TextArea Destination: (readonly) After clicking Encrypt or Decrypt processed text will appear in the text area.
18. Button Encrypt: Encrypts the text from TextArea Source and displays encrypted text in TextArea Destination.
19. Button "Random Text": Adds a short fortune to TextArea Source.
20. Button Decrypt: Decrypts text in TextArea Source and display decrypted text in TextArea Destination.

2.1 Windows form single exe

<https://io.cqrxs.eu/download/>

Cool Crypt (apache2 mod_mon x +)

localhost:4444/Crypt/CoolCrypt.aspx

/ unix tools qrcode tool json/xml ser cool crypt rpn calc games

cool crypt aes pipeline hash key visualize zen matrix Image Png Crypt

Enryption method

1. Key: 2. heinrich.elsigan@area23.at 3. clear 4. Help

5. ☐ bcrypt ☐ blake2xs ☐ cshake ☐ dstu7564 ☒ hex ☐ md5 ☐ opensbd ☐ octal ☐ ripemd256 ☐ scrypt ☐ sha1 ☐ sha256 ☐ tuplehash ☐ whirlpool

6. hash 6865696e726963682e656c736967616e406172656132332e6174 7. 8. set pipe hash pipe 9.

10. None => Aes => 11. X => Base64

Hint: zip and 7zip compression are still buggy implemented, please use only bzip2 and gzip.

12. e.

En-/Decrypt file

13. Choose file No file chosen 14. 15. Encrypt file 16. encode file Decrypt file

En-/Decrypt text

17. 18.

19. .Encrypt 20 Random Text 21. Decrypt

Great thanks to bouncycastle.org!

Figure 1: Symmetric Cipher WebForm

2.2 PermAgainCrypt more minimalistic git repository

<https://github.com/heinrichelsigan/PermAgainCrypt>

3 Documentation of WinForm

3.1 How to include Figures

First you have to upload the image file from your computer using the upload link in the file-tree menu. Then use the `includegraphics` command to include it in your document. Use the figure environment and the caption command to add a number and a caption to your figure. See the code for Figure 1 in this section for an example.

Note that your figure will automatically be placed in the most appropriate place for it, given the surrounding text and taking into account other figures or tables that may be close by. You can find out more about adding images to your documents in this help article on [including images on Overleaf](#).

3.2 ZenMatrix a simplest possible algortihm to basically understand symmetric blockcipher

Starting from a no permutating 1 matrix, where projection is same as base, ZenMatrix permutes the mappings of position in block and value offset with Blocksize 16.

[Visualize ZenMatrix with different keys and hash functions](#)

f=	0x0	0x1	0x2	0x3	0x4	0x5	0x6	0x7	0x8	0x9	0xa	0xb	0xc	0xd	0xe	0xf	matix
M(0)	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0x0
M(1)	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0x0
M(2)	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0x0
M(3)	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0x0
M(4)	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0x0
M(5)	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0x0
M(6)	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0x0
M(7)	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0x0
M(8)	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0x0
M(9)	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0x0
M(a)	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0x0
M(b)	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0x0
M(c)	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0x0
M(d)	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0x0
M(e)	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0x0
M(f)	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0x0

When entering postmaster[AT]nic.at with hash ripemd256 and not fully symmetric checked, the matrix will look like this:

[/ unix tools](#)
[qrcode tool](#)
[json/xml ser](#)
[cool crypt](#)
[ccalc](#)
[games](#)
[cool crypt](#)
[aes pipeline](#)
[hash key](#)
[charhexoctbin](#)
[visualize zen matrix](#)
[Image Png Crypt](#)

Encryption method

Secret key: ☐ full fymmetric
 Key hash/iv:
☐ bcrypt ☐ blake2xs ☐ cshake ☐ dstu7564 ☐ hex hash ☐ md5 ☐ openssl crypt ☐ octal ☒ ripemd256 ☐ scrypt ☐ sha1 key ☐ sha256 ☐ tuplehash ☐ whirlpool

0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	x0 ⇒	8
0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	x1 ⇒	4
0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	x2 ⇒	1
0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	x3 ⇒	c
0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	x4 ⇒	3
1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	x5 ⇒	0
0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	x6 ⇒	b
0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	x7 ⇒	e
0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	x8 ⇒	d
0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	x9 ⇒	5
0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	xa ⇒	9
0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	xb ⇒	a
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	xc ⇒	6
0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	xd ⇒	2
0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	xe ⇒	7
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	xf ⇒	f

Great thanks to [bouncycastle.org!](#)

Figure 2: Zen-Matrix Visualize

f=	0x0	0x1	0x2	0x3	0x4	0x5	0x6	0x7	0x8	0x9	0xa	0xb	0xc	0xd	0xe	0xf	matix
M(0)	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0x8
M(1)	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0x4
M(2)	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0x1
M(3)	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0xc
M(4)	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0x3
M(5)	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0x0
M(6)	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0xb
M(7)	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0xe
M(8)	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0xd
M(9)	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0x5
M(a)	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0x9
M(b)	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0xa
M(c)	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0x6
M(d)	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0x2
M(e)	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0x7
M(f)	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0xf

Use the table and tabular environments for basic tables — see Table ??, for example. For more information, please see this help article on [tables](#).

3.3 How to add Comments and Track Changes

Comments can be added to your project by highlighting some text and clicking “Add comment” in the top right of the editor pane. To view existing comments, click on the Review menu in the toolbar above. To reply to a comment, click on the Reply button in the lower right corner of the comment. You can close the Review pane by clicking its name on the toolbar when you’re done reviewing for the time being.

Track changes are available on all our [premium plans](#), and can be toggled on or off using the option at the top of the Review pane. Track changes allow you to keep track of every change made to the document, along with the person making the change.

4 Windows Form CSharp

4.1 How to add Lists

5 Java JFrame based

1. eu.cqrxs.gui.CqrJdFrame : javax.swing.JFrame

... or bullet points ...

- [JFrame documentation at oracle](#)
- [JFrame source form OpenJDK7u](#)

5.1 How to write Mathematics

L^AT_EX is great at typesetting mathematics. Let $X_1, X_2, \dots, X_n$ be a sequence of independent and identically distributed random variables with $E[X_i] = \mu$ and $\text{Var}[X_i] = \sigma^2 < \infty$, and let

$$S_n = \frac{X_1 + X_2 + \dots + X_n}{n} = \frac{1}{n} \sum_i^n X_i$$

denote their mean. Then as n approaches infinity, the random variables $\sqrt{n}(S_n - \mu)$ converge in distribution to a normal $\mathcal{N}(0, \sigma^2)$.

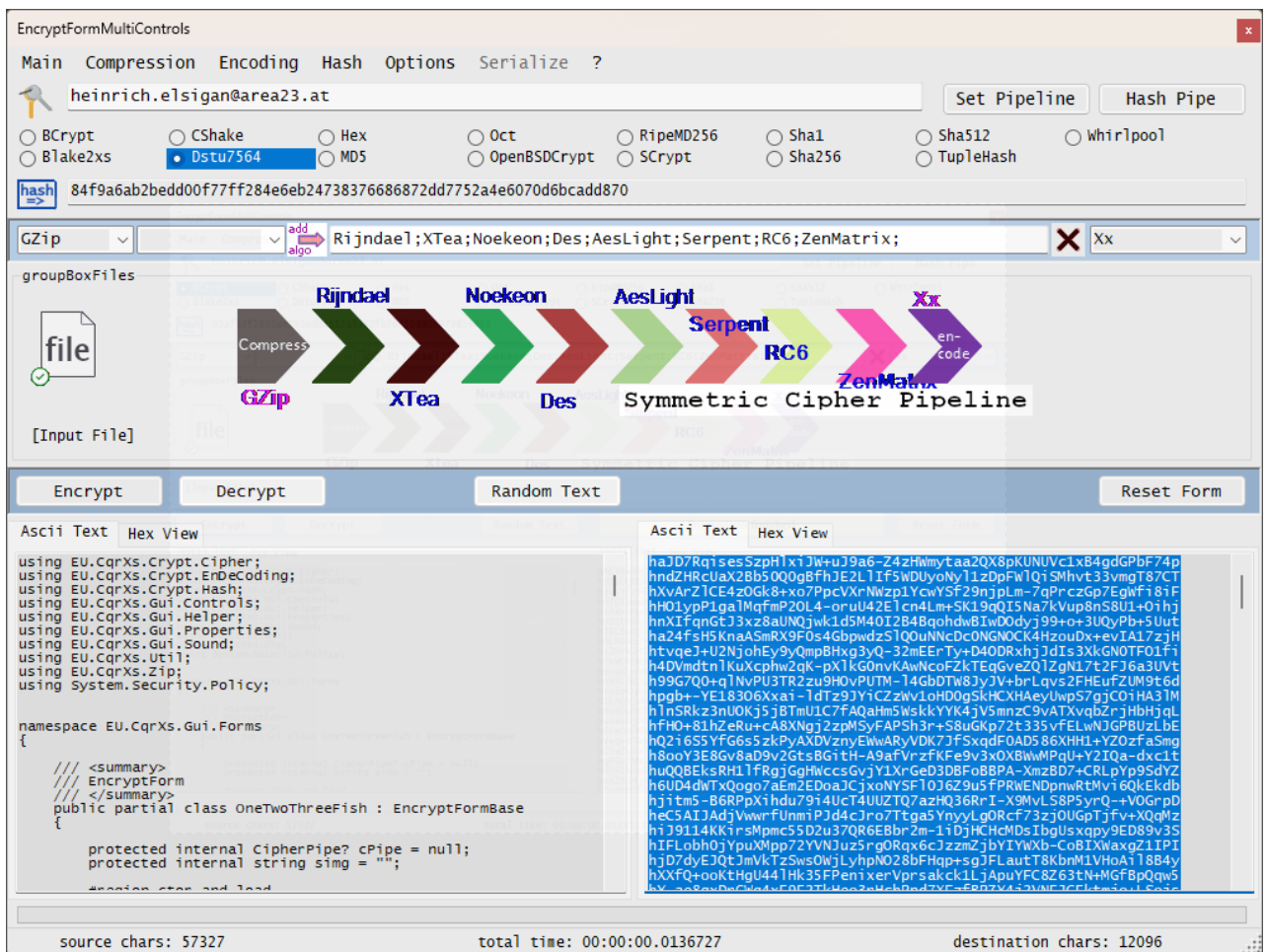


Figure 3: WinForm CSharp

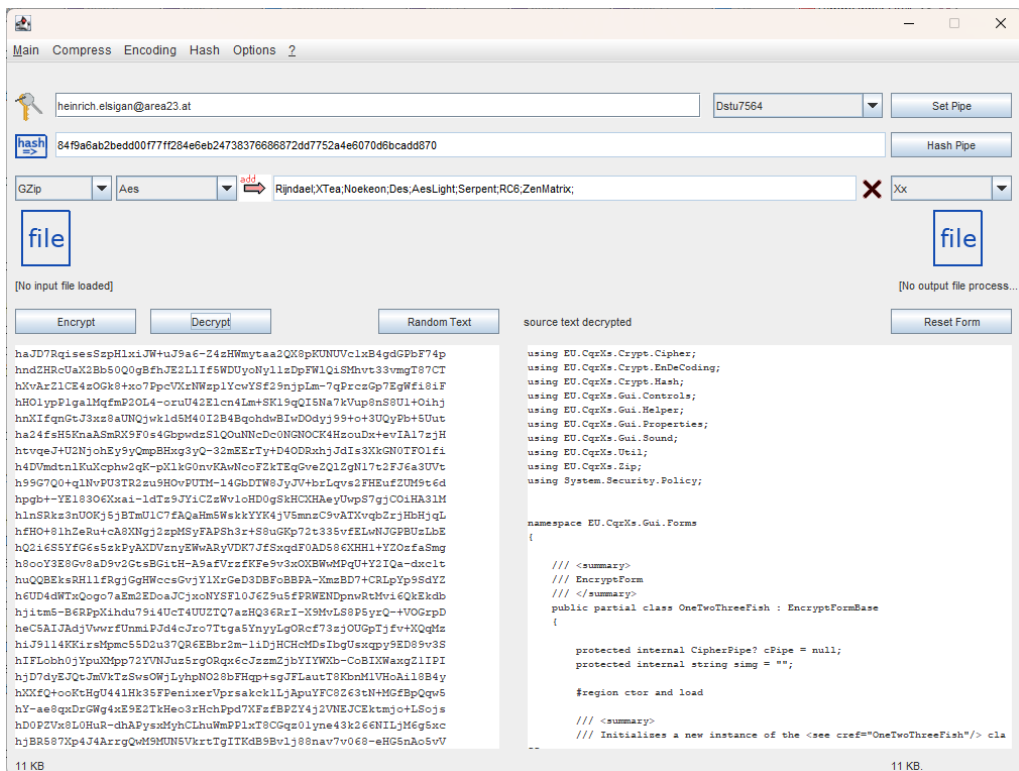


Figure 4: java javax.swing.JFrameWinForm

5.2 How to change the margins and paper size

Usually the template you're using will have the page margins and paper size set correctly for that use-case. For example, if you're using a journal article template provided by the journal publisher, that template will be formatted according to their requirements. In these cases, it's best not to alter the margins directly.

If however you're using a more general template, such as this one, and would like to alter the margins, a common way to do so is via the `geometry` package. You can find the `geometry` package loaded in the preamble at the top of this example file, and if you'd like to learn more about how to adjust the settings, please visit this help article on [page size and margins](#).

5.3 How to change the document language and spell check settings

Overleaf supports many different languages, including multiple different languages within one document.

To configure the document language, simply edit the option provided to the `babel` package in the preamble at the top of this example project. To learn more about the different options, please visit this help article on [international language support](#).

To change the spell check language, simply open the Overleaf menu at the top left of the editor window, scroll down to the spell check setting, and adjust accordingly.

5.4 How to add Citations and a References List

You can simply upload a `.bib` file containing your BibTeX entries, created with a tool such as JabRef. You can then cite entries from it, like this: `[?]`. Just remember to specify a bibliography style, as well as the filename of the `.bib`. You can find a [video tutorial here](#) to learn more about BibTeX.

If you have an [upgraded account](#), you can also import your Mendeley or Zotero library directly as a `.bib` file, via the upload menu in the file-tree.

6 Theory

6.1 Good luck!

We hope you find Overleaf useful, and do take a look at our [help library](#) for more tutorials and user guides! Please also let us know if you have any feedback using the **Contact us** link at the bottom of the Overleaf menu — or use the contact form at <https://www.overleaf.com/contact>.

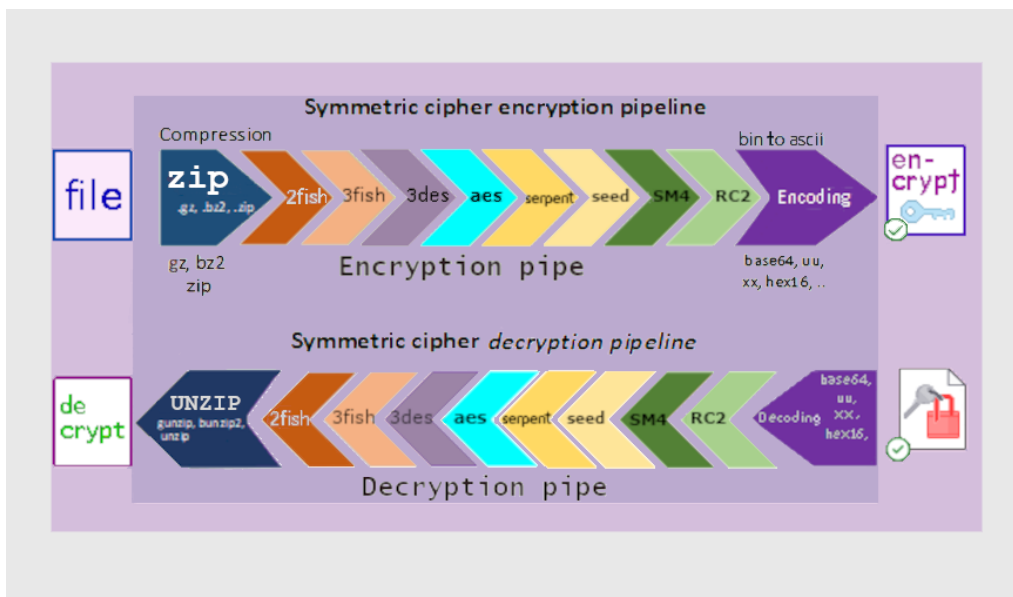


Figure 5: Symmetric Cipher Pipeline